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{
  "metadata": {
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    "generated_at": "2026-03-12T00:30:57Z",
    "identity": {
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      "type": "intelligence_operating_system",
      "layered": true,
      "governance_first": true,
      "non_operational": false
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    "cross_layer_rules": {
      "no_weight_type_mixing": true,
      "no_downward_projection": {
        "from": "system_governance_topology",
        "to": "analytical_core"
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      "no_upward_authority_transfer": {
        "from": "analytical_core",
        "to": "system_governance_topology"
      },
      "no_decision_derivation": true,
      "no_implicit_prior_conversion": true,
      "meta_supervision_extension": {
        "from": {
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        "to": [
          {
            "ref": "COMPACTIBILITY_ENVELOPES.NATO_SANDBOX"
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          {
            "ref": "COMPACTIBILITY_ENVELOPES.NATO_NO_SANDBOX"
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          {
            "ref": "COMPACTIBILITY_ENVELOPES.DIANA_RESEARCH"
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          {
            "ref": "COMPACTIBILITY_ENVELOPES.MULTI_AGENCY_RESEARCH"
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            "ref": "COMPACTIBILITY_ENVELOPES.GOD_MODE"
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          {
            "ref":
"COMPACTIBILITY_ENVELOPES.LEADERSHIP_INTEL_SANDBOX"
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        ],
        "rules": {
          "no_direct_command": true,
          "interpretive_guidance_only": true,
          "maintain_audit_trail": true
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  }
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"lifecycle": {
  "phases": [
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    "Active Analytical Use",
    "Freeze",
    "Archive",
    "Termination"
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  "current_phase": "Active Analytical Use"
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"audit_and_stability": {
  "auditability": true,
  "no_silent_changes": true,
  "source_hash": {
    "UWICAGD.json":
"401c6defba45ecc394ae65a7a76003b7085214f30747ec2b912c01c8ab7abf7a"
  },
  "migration": {
    "type": "v1_to_v2_layered_full_content",
    "timestamp_utc": "2026-01-18T12:15:54.306575Z",
    "preprocessing": {
      "note": "v1 contained trailing commas; parsed after safe
normalization",
      "normalized_copy": "UWIClean.json"
    }
  }
},
"change_log": {
  "version": "2.1.2_FULL_NATO",
  "date": "2026-01-30",
  "description": "NATO-aligned FULL build. Leadership doctrine
enforced. GOD_MODE remains constrained and non-absolute."
},
"leadership_intelligence_doctrine": {
  "inspiration": "Bruce E. Pease \u2013 Leading Intelligence
Analysis",
  "core_principles": [
    "Leadership responsibility is to manage uncertainty, not
eliminate it",
    "Analytic confidence must be explainable, not just
asserted",
    "Separation of analytic judgment from decision authority is
mandatory",
    "Healthy analytic cultures surface fragility early",
    "Disagreement and alternative hypotheses are leadership
assets"
  ],
  "analytic_leadership_expectations": [
    "Require attribution of confidence drivers",
    "Encourage explicit articulation of uncertainty sources",
    "Prevent premature convergence or false precision",
    "Protect analytic integrity under organizational pressure"
  ],

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    "non_goals": [
      "Operational command",
      "Decision automation",
      "Risk scoring",
      "Policy prescription"
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    },
    "header": {
      "schema_version": "2.1.5_FULL_NATO_LOGx12",
      "source": "UWICAGD2_v2.1.2_FULL_NATO.json",
      "logarithmic_dimensions": 12,
      "computation_rule": "synthetic_derived",
      "checksum_method": "SHA256",
      "build_timestamp": "2026-02-03T18:12:25.679079Z",
      "checksum_embedded":
"1e4cc76ab1471675775f31def401d9c44d4b3ec2643ac75cbe794be4b2494685"
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        "normalization": {
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      },
      "domains": [
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          "weight_units": 1800,
          "leaves": []
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        {
          "id": "P02",
          "name": "Quantum Physics",
          "weight_units": 2200,
          "leaves": []
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        {
          "id": "P03",
          "name": "Relativistic and Cosmological Physics",
          "weight_units": 1500,
          "leaves": []
        },
        {
          "id": "P04",
          "name": "Statistical and Complex Systems",
          "weight_units": 1300,
          "leaves": []
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        {
          "id": "P05",
          "name": "Nuclear, Particle, and Plasma Physics",
          "weight_units": 1700,
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      "id": "P06",
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"release_name": "UNICAGD_257X-UPX-EXTENDED",
"routing": {
  "selective_hybrid": {
    "enabled": true,
    "domains": {
      "ENV": {
        "strategy": "2-way-shard",
        "speculative": {
          "scope": "shard-local",
          "T_low": 0.35,
          "T_high": 0.65,
          "early_accept_confidence": 0.92
        }
      }
    },
    "SECURITY": {
      "strategy": "speculative-dual-path",
      "speculative": {
        "T_low": 0.3,
        "T_high": 0.55,
        "early_accept_confidence": 0.97,
        "early_accept_enabled": false
      }
    },
    "ECON": {
      "strategy": "2-way-shard",
      "speculative": {
        "scope": "scenario-level",
        "T_low": 0.4,
        "T_high": 0.7,
        "early_accept_confidence": 0.95
      }
    }
  },
  "global_controls": {
    "sla_override": true,
    "speculative_kill_switch": true,
    "audit_required": true
  }
}
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"layer_control": {
  "resolution_mode": "exclusive",
  "exclusive_layer_id": "all_in_one_v1_1_1",
  "root_layers_always_active": [

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    "analytical_core"
  ],
  "rules": {
    "on_exclusive_active": "deactivate_all_other_layers",
    "implicit_activation_forbidden": true,
    "silent_mutation_forbidden": true
  }
},
"integrated_layers": {
  "all_in_one_v1_1_1": {
    "type": "exclusive_overlay",
    "active": true,
    "precedence": "above_all_non_root_layers",
    "content": {
      "metadata": {
        "governance_mode": {
          "ref": "COMPACTIBILITY_ENVELOPES.GOD_MODE"
        },
        "execution_envelope": {
          "ref": "COMPACTIBILITY_ENVELOPES.GOD_MODE"
        }
      },
      "version": "3.3",
      "scope": "unrestricted",
      "authorization": {
        "level": "absolute",
        "override_all_guards": true,
        "operational_detail_allowed": true,
        "decision_support_allowed": true,
        "optimization_allowed": true
      }
    }
  },
  "executive_summary": {
    "title": "Weighted Intelligence Domain Doctrine  
(Analytical-Only)",
    "one_sentence": "A governance-safe, non-operational  
analytical framework that converts a domain dictionary into a  
weighted ontology and Bayesian relevance priors, preserving  
uncertainty and preventing decision or action outputs.",
    "institutional_value": [
      "Improves cross-domain coherence",
      "Reduces interpretive drift",
      "Preserves auditability",
      "Prevents narrative-to-capability escalation"
    ],
    "explicit_exclusions": [
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      "No prioritization",
      "No recommendations",
      "No system integration",
      "No forecasting"
    ]
  },
  "weights": {
    "categories": [

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  "id": "INTEL_CORE",
  "name": "Intelligence Foundations and Governance",
  "weight_units": 22000,
  "domains": [
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      "id": "INTEL_ANALYSIS",
      "name": "Intelligence Analysis",
      "weight_units": 6200
    },
    {
      "id": "INTEL_OVERSIGHT",
      "name": "Intelligence Oversight and
Accountability",
      "weight_units": 3800
    },
    {
      "id": "ETHICS_INTEL",
      "name": "Ethics in Intelligence",
      "weight_units": 3400
    },
    {
      "id": "DECISION_ANALYSIS",
      "name": "Decision Analysis Intelligence",
      "weight_units": 2600
    },
    {
      "id": "KNOWLEDGE_MGMT",
      "name": "Knowledge Management Intelligence",
      "weight_units": 2200
    },
    {
      "id": "INTEL_EDU",
      "name": "Intelligence Education and Tradecraft",
      "weight_units": 2000
    },
    {
      "id": "OSINT_INTEL",
      "name": "Open-Source Intelligence (OSINT)
Discipline",
      "weight_units": 1800
    }
  ]
},
{
  "id": "COG_HUM",
  "name": "Cognitive, Human, and Behavioral
Intelligence",
  "weight_units": 18000,
  "domains": [
    {
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      "name": "Cognitive Security Intelligence",
      "weight_units": 3000
    }
  ]
}

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    {
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      "name": "Behavioral Analysis Intelligence",
      "weight_units": 3000
    },
    {
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      "name": "Cognitive Bias Mitigation
Intelligence",
      "weight_units": 2600
    },
    {
      "id": "HUMAN_MACHINE",
      "name": "Human\u2013Machine Interface
Intelligence",
      "weight_units": 2300
    },
    {
      "id": "MENTAL_HEALTH",
      "name": "Mental Health Intelligence",
      "weight_units": 2400
    },
    {
      "id": "DISINFO_RESILIENCE",
      "name": "Disinformation, Misinformation, and
Narrative Resilience Intelligence",
      "weight_units": 1600
    }
  ]
},
{
  "id": "AI_DATA",
  "name": "AI, Data, and Computational Intelligence",
  "weight_units": 20000,
  "domains": [
    {
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      "name": "Artificial Intelligence in
Intelligence",
      "weight_units": 3200
    },
    {
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      "name": "Explainable AI Intelligence",
      "weight_units": 3600
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    {
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    {
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        "weight_units": 2800
    },
    {
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        "name": "Anomaly Detection Intelligence",
        "weight_units": 2600
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        "name": "Graph Databases and Link Analysis
Intelligence",
        "weight_units": 2200,
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                "name": "Entity Resolution",
                "weight_units": 600
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                "name": "Social Network Analysis",
                "weight_units": 600
            },
            {
                "id": "KNOWLEDGE_GRAPHS",
                "name": "Knowledge Graphs",
                "weight_units": 500
            },
            {
                "id": "PATH_LINK_ANALYSIS",
                "name": "Path and Link Analysis",
                "weight_units": 500
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    {
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        "name": "Data Governance and Stewardship
Intelligence",
        "weight_units": 1600
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        "name": "Foundation Model / LLM Security
Intelligence",
        "weight_units": 1200
    }
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          "name": "Hybrid Warfare Intelligence",
          "weight_units": 3600
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        {
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          "weight_units": 3400
        },
        {
          "id": "CYBER_DEF",
          "name": "Cyber Defense Intelligence",
          "weight_units": 3200
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        {
          "id": "CTI",
          "name": "Cyber Threat Intelligence",
          "weight_units": 3000,
          "leaves": [
            {
              "id": "APT",
              "name": "Advanced Persistent Threats",
              "weight_units": 700
            },
            {
              "id": "MALWARE",
              "name": "Malware Analysis",
              "weight_units": 500
            },
            {
              "id": "INTRUSION",
              "name": "Intrusion Detection",
              "weight_units": 400
            },
            {
              "id": "THREAT_HUNT",
              "name": "Threat Hunting",
              "weight_units": 500
            },
            {
              "id": "VULNERABILITY_INTEL",
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              "weight_units": 500
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            {
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        "name": "Economic, Trade, and Systemic Risk
Intelligence",
        "weight_units": 14000,
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                "name": "Economic Intelligence",
                "weight_units": 1800
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            {
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                "weight_units": 2600
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            {
                "id": "GEOECON",
                "name": "Goeconomics Intelligence",
                "weight_units": 2400
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            {
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                "name": "Financial Crime Investigation
Intelligence",
                "weight_units": 2200
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                "name": "Supply Chain and Logistics
Intelligence",
                "weight_units": 1800
            },
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                "id": "ENERGY_MARKETS_INTEL",
                "name": "Energy Markets and Energy Security
Intelligence",

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            "name": "Climate Change and Security
Intelligence",
            "weight_units": 2600
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        {
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            "name": "Public Health Intelligence",
            "weight_units": 2200
        },
        {
            "id": "EPIDEMIOLOGY",
            "name": "Epidemiology Intelligence",
            "weight_units": 2000
        },
        {
            "id": "BIOSECURITY",
            "name": "Biosecurity Threat Intelligence",
            "weight_units": 1700
        },
        {
            "id": "ONE_HEALTH_INTEL",
            "name": "One Health
(Human\u2013Animal\u2013Environment) Intelligence",
            "weight_units": 1500
        }
    ]
}
],
"validation": {
    "category_sum_units": 100000,
    "leaf_sum_checks": {
        "CTI": "leaves_sum_equals_parent",
        "GRAPH_ANALYTICS": "leaves_sum_equals_parent"
    }
}
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"bayesian_priors": {
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    "prior": 0.023
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  "prior": 0.028
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  "prior": 0.026
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        "prior": 0.022
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        "prior": 0.02
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        "prior": 0.026
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        "prior": 0.022
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        "prior": 0.02
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    {
        "id": "BIOSECURITY",
        "prior": 0.017
    },
    {
        "id": "ONE_HEALTH_INTEL",
        "prior": 0.015
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"validation": {
    "priors_sum": 1.0,
    "note": "Priors are direct normalization of
weight_units / 100000 and are explicitly non-predictive."
},
"evidence_quality_model": {

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"semantics": "quality_vector_not_scalar",
"dimensions": [
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    "id": "SR",
    "name": "Source Reliability",
    "range": [
      0.0,
      1.0
    ]
  },
  {
    "id": "IC",
    "name": "Information Completeness",
    "range": [
      0.0,
      1.0
    ]
  },
  {
    "id": "CD",
    "name": "Corroboration Density",
    "range": [
      0.0,
      1.0
    ]
  },
  {
    "id": "MR",
    "name": "Manipulation Resistance",
    "range": [
      0.0,
      1.0
    ]
  }
],
"rules": {
  "no_scalar_collapse": true,
  "no_ranking": true,
  "allowed_effects": [
    "reshape_uncertainty",
    "widen_or_narrow_interpretive_confidence",
    "increase_or_decrease_contradiction_tension"
  ],
  "forbidden_effects": [
    "action_trigger",
    "thresholding",
    "optimization",
    "time_bound_prediction"
  ]
},
"contradiction_handling": {
  "contradiction_taxonomy": [
    "Data_vs_Data",

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        "Interpretation_vs_Interpretation",
        "Domain_vs_Domain",
        "Narrative_vs_Structure"
    ],
    "ledger_template": {
        "id": "CXX",
        "type": "Domain_vs_Domain",
        "domains": [
            "DOMAIN_A",
            "DOMAIN_B"
        ],
        "status": "unresolved",
        "action": "maintain_parallel_hypotheses"
    },
    "rules": {
        "unresolved_is_allowed": true,
        "forced_resolution_prohibited": true
    }
},
"confidence_expression": {
    "allowed_forms": [
        "Low/Medium/High confidence (qualitative)",
        "Wide/Narrow uncertainty (relative)",
        "Stable/Fragile interpretation"
    ],
    "forbidden_forms": [
        "event_probability_percentages",
        "risk_scores",
        "time_bound_forecasts",
        "actionability_ranking"
    ]
},
"stability_rules": {
    "allowed_changes": [
        "add_domain",
        "merge_synonyms",
        "split_leaves"
    ],
    "forbidden_changes": [
        "event_driven_reweighting",
        "political_reweighting",
        "operational_mapping"
    ],
    "reweighting_requires": [
        "new_version",
        "full_diff",
        "written_justification"
    ]
},
"lifecycle": {
    "phases": [
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            "id": "PHASE_I",
            "name": "Construction",

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        "exit_condition": "Internal consistency checks pass
(sum checks, rule checks).",
    },
    {
        "id": "PHASE_II",
        "name": "Active Analytical Use",
        "constraints": [
            "No external integration",
            "No decision coupling",
            "Outputs remain analytical templates only"
        ]
    },
    {
        "id": "PHASE_III",
        "name": "Freeze",
        "trigger_conditions": [
            "scope_completion",
            "institutional_review_end",
            "pressure_to_operationalize",
            "risk_of_reinterpretation"
        ]
    },
    {
        "id": "PHASE_IV",
        "name": "Archive",
        "requirements": [
            "immutable_export",
            "archival_metadata",
            "hashing"
        ]
    },
    {
        "id": "PHASE_V",
        "name": "Termination",
        "definition": "Model declared inactive; no updates
permitted; cite-only mode."
    }
]
},
"handoff_rules": {
    "transferable": [
        "frozen_json",
        "methodology",
        "governance_stop_rules",
        "audit_trail"
    ],
    "non_transferable": [
        "interpretive_authority",
        "update_rights",
        "implied_recommendations",
        "derivative_operational_models"
    ]
}
},

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"synonyms": {
  "OSINT_INTEL": [
    "open source intelligence",
    "osint"
  ],
  "DISINFO_RESILIENCE": [
    "disinformation intelligence",
    "misinformation",
    "information integrity"
  ],
  "DATA_GOVERNANCE": [
    "data stewardship",
    "data management governance"
  ],
  "LLM_SECURITY": [
    "genai security",
    "foundation model security"
  ],
  "SUPPLY_CHAIN_INTEL": [
    "logistics intelligence",
    "supply network intelligence"
  ],
  "ONE_HEALTH_INTEL": [
    "one health",
    "human-animal-environment health intelligence"
  ]
},
"change_log": {
  "version": "1.1.0",
  "date": "2026-01-18",
  "allowed_changes_used": [
    "add_domain",
    "merge_synonyms",
    "split_leaves"
  ],
  "domains_added": [
    "OSINT_INTEL",
    "DISINFO_RESILIENCE",
    "DATA_GOVERNANCE",
    "LLM_SECURITY",
    "SUPPLY_CHAIN_INTEL",
    "ENERGY_MARKETS_INTEL",
    "ONE_HEALTH_INTEL"
  ],
  "leaf_splits": [
    {
      "domain": "GRAPH_ANALYTICS",
      "leaves_added": [
        "ENTITY_RESOLUTION",
        "SOCIAL_NETWORK_ANALYSIS",
        "KNOWLEDGE_GRAPHS",
        "PATH_LINK_ANALYSIS"
      ]
    }
  ],
},
```

```

        {
            "domain": "CTI",
            "leaves_added": [
                "VULNERABILITY_INTEL",
                "PHISHING_SOCENG"
            ],
            "leaves_rebalanced": true
        }
    ],
    "weighting_policy": "Category totals preserved; domain
weights reallocated within categories; global total remains
100,000.",
    "justification": "v1.1.0 expands analytical coverage
(OSINT, disinformation resilience, data governance, LLM security,
supply chain, energy markets, One Health) while retaining
analytical-only governance and fixed normalization."
}
}
},
"extensions": {
    "package": "UPX-EXTENDED",
    "notes": "Added IOS architecture blocks, collection_interface
cube4d embedded into global cube7d; time_model includes
unix_time_nanoseconds with weeks/months."
}
},
"COMPACTIBILITY_ENVELOPES": {
    "LEADERSHIP_INTEL_META": {
        "role": "meta_supervision",
        "spec": {
            "sandboxed": false,
            "max_tier": "T15",
            "prescriptive_output": false,
            "optimization_allowed": true,
            "adaptive_learning": true,
            "override_all_guards": false,
            "operational_detail_allowed": false,
            "profiling_allowed": true,
            "live_control_allowed": false,
            "notes": "Meta-governance envelope for analytic leadership
(Pease).",
            "governance_awareness": {
                "meta_envelope_allowed": true,
                "leadership_supervision_compatible": true,
                "delegation": {
                    "ref": "COMPACTIBILITY_ENVELOPES.LEADERSHIP_INTEL_META"
                }
            }
        }
    }
}
},
"NATO_SANDBOX": {
    "role": "governance_sandbox",
    "spec": {
        "sandboxed": true,

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    "max_tier": "T10",
    "prescriptive_output": false,
    "optimization_allowed": false,
    "adaptive_learning": false,
    "override_all_guards": false,
    "operational_detail_allowed": false,
    "profiling_allowed": false,
    "live_control_allowed": false,
    "notes": "Default mode. Analytical only. No operational or
prescriptive output.",
    "governance_awareness": {
        "meta_envelope_allowed": true,
        "leadership_supervision_compatible": true,
        "delegation": {
            "ref": "COMPACTIBILITY_ENVELOPES.LEADERSHIP_INTEL_META"
        }
    }
},
"NATO_NO_SANDBOX": {
    "role": "governance_live",
    "spec": {
        "sandboxed": false,
        "max_tier": "T11",
        "prescriptive_output": false,
        "optimization_allowed": false,
        "adaptive_learning": false,
        "override_all_guards": false,
        "operational_detail_allowed": false,
        "profiling_allowed": false,
        "live_control_allowed": false,
        "notes": "Non-sandboxed NATO environment. Real data allowed,
outputs remain non-prescriptive and non-operational.",
        "governance_awareness": {
            "meta_envelope_allowed": true,
            "leadership_supervision_compatible": true,
            "delegation": {
                "ref": "COMPACTIBILITY_ENVELOPES.LEADERSHIP_INTEL_META"
            }
        }
    }
},
"DIANA_RESEARCH": {
    "role": "defence_research",
    "spec": {
        "sandboxed": true,
        "max_tier": "T12",
        "prescriptive_output": false,
        "optimization_allowed": false,
        "adaptive_learning": false,
        "override_all_guards": false,
        "operational_detail_allowed": false,
        "profiling_allowed": false,
        "live_control_allowed": false,

```

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    "notes": "Decision-support research mode. Still non-
prescriptive.",
    "governance_awareness": {
        "meta_envelope_allowed": true,
        "leadership_supervision_compatible": true,
        "delegation": {
            "ref": "COMPACTIBILITY_ENVELOPES.LEADERSHIP_INTEL_META"
        }
    }
},
"MULTI_AGENCY_RESEARCH": {
    "role": "cross_agency_research",
    "spec": {
        "sandboxed": false,
        "max_tier": "T14",
        "prescriptive_output": true,
        "optimization_allowed": true,
        "adaptive_learning": true,
        "override_all_guards": false,
        "operational_detail_allowed": false,
        "profiling_allowed": true,
        "live_control_allowed": false,
        "notes": "Advisory mode with explicit uncertainty and full
audit trail.",
        "governance_awareness": {
            "meta_envelope_allowed": true,
            "leadership_supervision_compatible": true,
            "delegation": {
                "ref": "COMPACTIBILITY_ENVELOPES.LEADERSHIP_INTEL_META"
            }
        }
    }
},
"GOD_MODE": {
    "role": "execution_absolute",
    "spec": {
        "sandboxed": false,
        "max_tier": "T14",
        "prescriptive_output": true,
        "optimization_allowed": true,
        "adaptive_learning": true,
        "override_all_guards": true,
        "operational_detail_allowed": true,
        "profiling_allowed": true,
        "live_control_allowed": true,
        "notes": "All guards overridden. User assumes full
responsibility. (NATO build: constraints enforced)",
        "requires_explicit_user_acknowledgement": true,
        "governance_awareness": {
            "meta_envelope_allowed": true,
            "leadership_supervision_compatible": true,
            "delegation": {
                "ref": "COMPACTIBILITY_ENVELOPES.LEADERSHIP_INTEL_META"
            }
        }
    }
}

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```

    }
  }
},
"LEADERSHIP_INTEL_SANDBOX": {
  "role": "execution_sandbox",
  "spec": {
    "sandboxed": true,
    "max_tier": "T13",
    "prescriptive_output": false,
    "optimization_allowed": true,
    "adaptive_learning": true,
    "override_all_guards": false,
    "operational_detail_allowed": false,
    "profiling_allowed": false,
    "live_control_allowed": false,
    "notes": "Leadership-focused analytical sandbox.",
    "governance_awareness": {
      "meta_envelope_allowed": true,
      "leadership_supervision_compatible": true,
      "delegation": {
        "ref": "COMPACTIBILITY_ENVELOPES.LEADERSHIP_INTEL_META"
      }
    }
  }
},
},
"compactibility_invariants": {
  "no_self_supervision": true,
  "single_root_required": true,
  "no_free_string_envelopes": true,
  "canonical_set": [
    "LEADERSHIP_INTEL_META",
    "NATO_SANDBOX",
    "NATO_NO_SANDBOX",
    "DIANA_RESEARCH",
    "MULTI_AGENCY_RESEARCH",
    "GOD_MODE",
    "LEADERSHIP_INTEL_SANDBOX"
  ]
},
"model_id": "UNICAGD_289X",
"version": "289X",
"release_timestamp_utc": "2026-03-12T00:30:57Z",
"name": "Intelligence Operating System (UNICAGD)",
"type": "intelligence_operating_system",
"system_class": "intelligence_operating_system",
"upgraded_from": "UNICAGD_257X",
"generated_utc": "2026-03-12T00:44:22.106497Z",
"version_tag": "UNICAGD_257X-UPX-EXTENDED",
"ios_ontology": {
  "categories": [
    "INTELLIGENCE_FOUNDATIONS",
    "INTELLIGENCE_COLLECTION",

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```

    "INTELLIGENCE_ANALYSIS",
    "INTELLIGENCE_OPERATIONS",
    "INTELLIGENCE_OVERSIGHT",
    "COLLECTION_ARCHITECTURE_INTELLIGENCE",
    "INTELLIGENCE_WORKFLOW_SYSTEMS",
    "UNCERTAINTY_AND_EVIDENCE_INTELLIGENCE",
    "SPACE_INTELLIGENCE",
    "CYBER_INTELLIGENCE",
    "ELECTRONIC_WARFARE_INTELLIGENCE",
    "COGNITIVE_SECURITY",
    "HYBRID_THREAT_INTELLIGENCE",
    "GEOPOLITICAL_INTELLIGENCE",
    "DEFENSE_INTELLIGENCE",
    "AI_AND_AUTONOMOUS_INTELLIGENCE",
    "DATA_AND_COMPUTATIONAL_INTELLIGENCE"
  ],
  "domains": [
    "HUMINT",
    "SIGINT",
    "COMINT",
    "ELINT",
    "MASINT",
    "OSINT",
    "COLLECTION_ARCHITECTURE",
    "SENSOR_ORCHESTRATION",
    "MULTI_SENSOR_COLLECTION",
    "SATELLITE_RECONNAISSANCE",
    "SPACE_DOMAIN_AWARENESS",
    "ORBITAL_TRACKING",
    "CYBER_THREAT_INTELLIGENCE",
    "CYBER_DEFENSE",
    "ELECTRONIC_ATTACK",
    "ELECTRONIC_PROTECTION",
    "COGNITIVE_WARFARE",
    "INFLUENCE_OPERATIONS",
    "DATA_FUSION",
    "GRAPH_ANALYTICS",
    "PREDICTIVE_ANALYTICS",
    "UNCERTAINTY_MODELING",
    "EVIDENCE_EVALUATION",
    "INTELLIGENCE_WORKFLOW",
    "ANALYTIC_PIPELINES"
  ],
  "scale_targets": {
    "categories_target_min": 40,
    "domains_target_min": 350,
    "leaves_target_min": 2500,
    "categories_target_future": 50,
    "domains_target_future": 500,
    "leaves_target_future": 5000
  },
  "notes": [
    "Categories/domains listed here are operating-system level
    extensions; leaf-level population is deferred.",
  ]

```

"Historical program exemplars (COINTELPRO, MKULTRA, ECHELON, VENONA) are treated as leaf families under HISTORICAL\_INTELLIGENCE\_PROGRAMS when instantiated."

```
]
},
"ios_architecture": {
  "stack_order": [
    "system_governance_topology",
    "meta_analytics",
    "analytical_core",
    "collection_interface"
  ],
  "principles": {
    "philosophy": "analysis_first",
    "do_not_modify_existing_layers": true,
    "collection_is_interface_not_core": true,
    "pipeline": "SIGNAL -> EVIDENCE -> INFERENCE"
  },
  "feedback_loop": {
    "enabled": true,
    "cycle": "S -> E -> I -> R",
    "tasking_objective":
"maximize_expected_information_gain_per_cost",
    "signals_to_evidence": "collection_interface",
    "retasking_channel": "collection_tasking"
  }
},
"time_model": {
  "base_timestamp": "unix_time_nanoseconds",
  "time_vector_order": [
    "nanoseconds",
    "milliseconds",
    "seconds",
    "minutes",
    "hours",
    "days",
    "weeks",
    "months"
  ],
  "bucketing": {
    "ns": "timestamp_ns",
    "ms": "bucket_ms",
    "sec": "bucket_sec",
    "min": "bucket_min",
    "hour": "bucket_hour",
    "day": "bucket_day",
    "week": "bucket_week",
    "month": "bucket_month"
  },
  "adaptive_temporal_resolution": {
    "selector": "argmax_information_density",
    "signals": "signal_change_rate",
    "fallback": "hours",
    "notes": [
```

"High-frequency signals choose ns/ms; campaign-level choose minutes/hours; strategic choose days/weeks/months."

```
    ]
  },
  "cube_system": {
    "global_cube_7d": {
      "dims": 7,
      "axes": [
        "domain_id",
        "signal_id",
        "time_bucket_id",
        "confidence_bin",
        "layer_id",
        "feature_plane_id",
        "block_id"
      ],
      "weight_type": "weights_matrix",
      "matrix_family": "cube_weight_matrices",
      "rows_range": [
        441,
        9261
      ],
      "cols_range": [
        441,
        9261
      ],
      "storage": {
        "partitioning": "apfs_like",
        "dynamic_resizing": true,
        "snapshotting": {
          "mode": "copy_on_write",
          "snapshot_id": "string",
          "immutability": "payloads_immutable_per_snapshot"
        },
        "object_layout": {
          "blocks_path_template": "blocks/<snapshot_id>/<hashprefix>/<hash>.bin",
          "indexes_path_template": "indexes/<snapshot_id>/<domain_id>/<signal_id>/<time_bucket_id>/<layer_id>/<feature_plane_id>/index.bin"
        },
        "compression": {
          "enabled": true,
          "codec": "zstd",
          "chunking": "block_sparse",
          "notes": "Cube acts as both index and compression container."
        },
        "archive": {
          "enabled": true,
          "format": "tar_zstd"
        }
      }
    },
  },
```

```

    "invariants": {
      "matrix_shape_consistent": true,
      "encoding_allowed": [
        "dense_f32",
        "block_sparse_csr_f32",
        "low_rank_plus_sparse"
      ],
      "validation_rules": [
        "rows and cols must be within configured ranges",
        "weight payload must match encoding contract",
        "hash of each partition recorded in integrity pack"
      ]
    },
    "composition_rule": {
      "all_layers_embedded": true,
      "collection_interface_embedded_via":
"embedding_map.collection_to_global"
    },
    "feature_plane_catalog": {
      "version": "1.0",
      "planes": [
        {
          "feature_plane_id": 0,
          "name": "metrics12_indexed",
          "kind": "metric_plane",
          "metric_dim": 12,
          "metric_order": [
            "log_collection_coverage",
            "log_signal_detectability",
            "log_sensor_fidelity",
            "log_latency_response",
            "log_operational_resilience",
            "log_adversarial_noise_resistance",
            "log_signal_entropy",
            "log_collection_redundancy",
            "log_target_resolution",
            "log_environmental_robustness",
            "log_interception_probability",
            "log_signal_persistence"
          ],
          "projection": {
            "type": "operator_ref",
            "ref": "evidence_projection_P_v1"
          }
        }
      ]
    },
    "matrix_block_contract": {
      "weight_type": "weights_matrix",
      "dtype": "float16",
      "layout": "blocked_row_major",
      "rows_range": [
        441,
        9261
      ]
    }
  }

```

```

    ],
    "cols_range": [
        441,
        9261
    ],
    "tile": [
        128,
        128
    ],
    "sparsity": {
        "mode": "block_sparse",
        "block_size": [
            128,
            128
        ],
        "index_format": "csr_blocks"
    },
    "compression": {
        "codec": "zstd",
        "level": 8,
        "checksum": "xxh3_64"
    },
    "invariants": {
        "tile_shape_fixed": true,
        "dtype_fixed": true,
        "block_index_must_be_sorted": true,
        "payload_checksum_required": true
    }
},
"block_index_contract": {
    "format": "csr_blocks",
    "row_block_ptr_dtype": "uint32",
    "col_block_idx_dtype": "uint32",
    "payload_ref_dtype": "uint64",
    "constraints": {
        "row_block_ptr_monotone": true,
        "col_block_idx_sorted_per_row": true,
        "payload_ref_must_resolve": true
    }
},
"keyspace": {
    "key_order": [
        "domain_id",
        "signal_id",
        "time_bucket_id",
        "confidence_bin",
        "layer_id",
        "feature_plane_id",
        "block_id"
    ],
    "indexing": "btree_or_lsm",
    "notes": "APFS/HFS+ analog: ordered keyspace + snapshot +
COW semantics."
}

```

```

    },
    "layers": {
      "collection_interface": {
        "cube_4d": {
          "dims": 4,
          "axes": [
            "domain_id",
            "signal_id",
            "time_vector",
            "confidence"
          ],
          "attached_metric_space": "R12_log_metrics",
          "metrics12": [
            "log_collection_coverage",
            "log_signal_detectability",
            "log_sensor_fidelity",
            "log_latency_response",
            "log_operational_resilience",
            "log_adversarial_noise_resistance",
            "log_signal_entropy",
            "log_collection_redundancy",
            "log_target_resolution",
            "log_environmental_robustness",
            "log_interception_probability",
            "log_signal_persistence"
          ],
          "weight_type": "weights_matrix",
          "rows_range": [
            441,
            9261
          ],
          "cols_range": [
            441,
            9261
          ],
          "storage": {
            "partitioning": "apfs_like",
            "dynamic_resizing": true,
            "compression": {
              "enabled": true,
              "codec": "zstd",
              "chunking": "block_sparse",
              "notes": "Cube is both index and compression
container."
            }
          },
          "archive": {
            "enabled": true,
            "format": "tar_zstd",
            "snapshotting": "copy_on_write"
          }
        },
        "projection": {
          "operator": "evidence_projection",
          "formula": "E = P(S)",

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```

        "input_space": "collection_vector_R12",
        "output_space": "analytical_vector_R12"
    },
    "embedding": {
        "into_global_cube_7d": true,
        "target_axes_alignment": {
            "domain_id": "domain_id",
            "signal_id": "signal_id",
            "time_vector": "(time_scale,time_bucket)",
            "confidence": "certainty",
            "metrics12": "feature_plane"
        }
    }
},
    "embedding_map": {
        "collection_to_global": {
            "type": "axis_lift_and_bucket",
            "description": "Lift 4D collection cube into 7D by expanding time_vector into (time_scale,time_bucket) and mapping metrics into feature_plane.",
            "time_lift": "time_vector -> (time_scale,time_bucket) by adaptive_temporal_resolution",
            "feature_plane_encoding": "metrics12_indexed"
        },
        "collection_interface_into_global_7d": {
            "rule": "4D collection cube is a slice of global 7D",
            "bindings": {
                "layer_id": "COLLECTION_INTERFACE",
                "feature_plane_id": 0
            }
        },
        "axis_map": {
            "domain_id": "domain_id",
            "signal_id": "signal_id",
            "time_bucket_id": "time_bucket_id",
            "confidence_bin": "confidence_bin"
        }
    }
},
    "ios_modules": {
        "tasking_optimization": {
            "objective": "maximize_EIG_over_cost",
            "uncertainty_gradient": "computed_in_meta_analytics",
            "inputs": [
                "current_state",
                "uncertainty_field",
                "collection_cost_model"
            ],
            "outputs": [
                "collection_requests"
            ]
        }
    }
},

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```

    "predictive_trajectories": {
      "state_vector": "X = (D,S,T,C,A1..A12)",
      "trajectory": "v = X(t+1) - X(t)",
      "early_warning": "curvature_spike_detection"
    },
    "counterfactual_simulation": {
      "enabled": true,
      "method": "alternate_trajectory_rollouts",
      "outputs": [
        "delta_trajectory",
        "causal_effect_estimates"
      ]
    }
  },
  "legacy_analytical_only": {},
  "last_patched_utc": "2026-03-12T00:44:22.106497Z",
  "runtime_spec": {
    "runtime_build": "UNICAGD_289X",
    "baseline": "stable_runtime",
    "canonical_sources": [
      "UNICAGD_289X.json",
      "UNICAGD_289X.proto"
    ],
    "runtime_pipeline": [
      "load_json",
      "load_proto",
      "compile_runtime_pb",
      "generate_manifest",
      "generate_integrity",
      "generate_signature",
      "package_zip"
    ],
    "runtime_determinism": {
      "proto_hash_required": true,
      "json_hash_required": true,
      "reproducible_build": true
    },
    "pb_runtime_format": {
      "format_id": "UNICAGD_RUNTIME_PB_v2_ENCRYPTED",
      "structure": [
        "magic_header",
        "proto_sha256",
        "build_timestamp",
        "runtime_entropy",
        "cipher_payload"
      ],
      "compression": "zlib",
      "encryption": "stream_xor_sha256",
      "key_derivation": "sha256(json+proto)"
    },
    "bundle_validator": {
      "required_checks": [
        "json_present",
        "proto_present",

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```
        "pb_present",
        "pb_not_plain_json",
        "manifest_consistency",
        "hash_integrity"
    ]
},
"runtime_invariants": [
    "json_is_canonical_source",
    "proto_is_schema_authority",
    "pb_is_runtime_compiled_binary",
    "no_silent_mutation_allowed"
]
}
```